Research Assignment 1

Section A

① A database is a collection of data or information stor
electronically in a computer
- databases help us store information due to large amoun
of data being generated everyday.
- Inventory tracking (online Shopping)
- Social media, managing user profiles and messaging data.

② A centralized storage used to collect, integrate and
manage large amounts of data from various business sour
- A regular database stores data from 1 source while
data warehouse stores data from multiple sources.
- large business need data warehouse to store cleaned versio
of various data from various departments into one large da
warehouse for simpler analysis.

③ A large centralized database that stores data in its
raw uncleaned form.
- Data stored in a warehouse has to be cleaned, while d
stored in a data lake is raw and unclean.
- When a business wants the data raw and not cleaned.

④ Data mart is a subset of a data warehouse and it s
data that is tailored to a single department.
- Managers and analysts ~~of~~ within a specific department.

⑤ Structured data conforms to a structure, it can
into a rigid / tables so it has rows and columns. Semi-str
data contains internal markers, Unstructured has no structure.

⑥ data modelling is the process of creating a database, it defines how data is collected, stored, structured and updated
- Ensures data quality
- Reduces costs.
- Reduces redundancy

⑦ Data mining is the process of extracting meaningful insig from massive datasets to identify hidden patterns and correlations can predict future trends.
- Businesses use data mining to extract meaning insights to make business decisions.
- youtube ~~use~~ showing recommendations after swiping through.
- Google ads from previous searches.

⑧ DBMS is a software used to create, store and manag databases.
① mysql is used for websites and business applications becaus it is fast and easy to use.
② postgreSQL is used for large systems that need strong security and advanced features.
③ ORacle is used by large companies like FMCG companies to handle large amounts of data.
④ MongoDB is used for apps that store flexible and unstructur data ~~ete~~

⑨ ETL means Extract, Transform and Load.
Extract means taking data from ~~p~~different places.
Transform means cleaning the data into the correct format
Load means loading the data into a database.
- ETL is important because it helps companies use clean and organised data for reports and decisions.

(10) CSV - Stores data in rows and columns, it is simple and works well in excel

Parquet ~~JSON~~ - Stores large amounts of data in a compressed way, it is good for big data analytics.

JSON ~~Parquet~~ - Stores data in key-value format, used a lot with A

Avro - Stores data in a compact format, useful when data is moved between systems.

- Use CSV for simple tables, JSON for API's, parquet for big data and Avro for fast data transfer.

Section B

(11) AI means ~~to~~ making computers do tasks that normally req human intelligence.

- Narrow AI does one specific task e.g Siri or chatgpt.
- General AI thinks and learn like a human.

(12) Machine learning is when we teach computers or feed com large amounts of data to learn from.

- Traditional programming uses rules written by people while m learns pattern from data.

Supervised learning learns, from labelled data.
Unsupervised learning, machine finds patterns in data without labels.
Reinforcement learning, learns by rewards and mistakes.

(13) Deep learning is a type of ML that uses neural networks.

- It is part of ML and ML is part of AI
- Neural network work like a simple version of the human bra They learn pattern from data and improve overtime.

(14) Mean Natural Language processing.
- It helps computers understand human language.
- Google translate, chatbots, Voice assistants.

(15) Large Language Model is an AI that is trained on large am
of text.
- It differs from normal AI because it can understand and
generate human-like language.
- Grok, used for answering questions and writing
- Claude, used for writing, research and summarising.

(16) Gen AI creates new content like text, images, music or code.
- It works by learning patterns from data and then creating
Something new.
Chatgpt creates text
DALL-E creates images
Copilot helps write code.

(17) Training dataset is the data used to teach an AI model
- Good data is important because the machine learns from it.
- If the data is wrong or biased, the AI can give unfair,
wrong or harmful results.

(18) Computer vision helps computers understand images and vid
- Computers learn/see by analysing pixels and patterns in images
- face recognition on phones.
- Self driving cars detecting roads and objects.

Section C

(19) An API is a way for two software systems to talk to ea other.
- A waiter in a restaurant, you tell the waiter your order and the waiter sends it to the kitchen and bring food back.
- They are important because they allow systems to share data.

(20) An API key is like a password for using an API
- It identifies who is using the API
- It must be kept private and not shared publicly
- If exposed, someone can misuse it and steal your data wh can be harmful, sometimes costly.

(21) Representational State Transfer.
- A REST API allows systems to communicate using HTTP.

GET: Read data
POST: Add new data
PUT: Update existing data
DELETE: Remove data.

(22) Javascript Object Notation.
- Used to store and send data in a simple format.

```
{
  "name": "Dakalo",
  "course": "Data analytics",
  "Age": 25
```

- It is common because its easy for computers and huma to read.

(23) Means writing clear instructions for AI tools
- It is important because when you give clear prompts you g
clear and better answers.
poor prompt: Tell me about data
Better prompt: Explain data analytics in simple words and give 2
real world examples.
poor prompt: Write something about AI
Better prompt: Write a short paragraph explaining AI and giv
the history of it and how far we are now.

(24) A map function is used to apply the same action to every
in a list.
- It saves time because you do not have to change each item
one by one

```
numbers = [1, 2, 3, 4]
doubled_numbers = list(map(lambda x: x * 2, numbers))
print(doubled_numbers)
```

(25) Cloud computing is when you use internet-based services,
Storage, software and computing power.
Iaas: Infrastructure as a Service, Amazon EC2
Paas: platform as a service, Google App engine.
Saas: Software as a service, google workspace.

(26) A version control System tracks changes in files and code.
- Git is the most popular version control tool
- Data analysts and developers use Git to save work, track change
and work with others.
Commit: A saved change
Branch: A Separate version of a project.

(27) A tool used to write code, notes and show results in one place.
- mostly used with python.
- Data analysts and Scientists like it because it is good for data analysis, testing code and showing charts.

(28) It is a popular programming language.
- It is popular in data analytics and AI because it has a huge community, easy to learn and has many useful libraries
pandas: Used for data tables.
Numpy: Used for numbers and Calculations.
Matplotlib: Used for charts.
Scikit-learn: Used for machine learning.

Section D

(29) Data privacy means protecting people's personal information.
- It is important because people have a right to control their data
- GDPR means the General Data Protection Regulation. It's a law that protects personal data in a European Union and European Economic area. It ~~is a~~ also affect companies outside europe if they handle EU people's data.

(30) Data ethics means using data in a fair and responsible way.
(1) Using people's data without permission
(2) Selling private data without telling users.
(3) Using biased AI that treats people unfairly
- Wrong because they can harm people and break trust.

(31) Managing data properly in an organization, it makes sure is accurate, safe and unused correctly.
- Data quality
- Data Security
- Data privacy
- Data ownership
- Rules and policies.

(32) Means AI giving unfair results because of the data it w trained on.
* Example
- A hiring AI may reject more women if it was trained mostly past male employee data.
- Causes unfair treatment and discrimination.

(33) Role	what they do	Skills	Tools.
Data analyst	Studies data and Create reports	Excel, SQL dashboards	Excel, power BI Tablrau
Data Scientist	Builds models and make predictions	python, Statistics ML	python, Jupyter, Scikit-learn.
Data Engineer	Builds data Systems and pipelines	SQL, databases ETL	SQL, Spark Airflow
AI Engineer	Builds AI Systems	ML, Deep learning	python, tensorflow, pytor

(34) BI means using data to help businesses make better dec
(1) power BI: Creates dashboards and reports.
(2) Tableau: Help visualise and analyse data.
- BI helps businesses ~~to~~ see trends, performance and problems.

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Section E.

(35) The concept i found most interesting is AI and ML. I like how AI and ML can help businesses analyse dat make predictions and automate tasks. It was interesting to learn how computers can learn from data and improve over time.

(36) Most difficult was deep learning and neural networks, at firs it was confusing to understand how computers learn using neural networks. I used Youtube videos and google search to understand it better. Simple examples and diagrams helped me learn faster.

(37) I believe data and AI will help me a lot in my future career. I want to work in data analytics and business operat so AI and data can help me make better decisions, analyze busi performance, reduce costs and improve efficiency.
- In everyday life AI can help with things like online assistant recommendations, navigation apps and faster access to information.

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